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Perspective: Acoustic Metamaterials in Future Engineering

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ABSTRACT

Acoustic metamaterials (AMMs) are a type of artificial materials that make use of appropriate structural designs and exhibit exotic properties not found in natural materials, such as negative effective material parameters (e.g., bulk modulus, mass density, and refractive index). These interesting properties offer novel means for sound manipulation and thus have drawn a great deal of attention. Over the past two decades, tremendous progress has been made in the fundamental research of AMMs, which has not only promoted the development of modern acoustics but also shown the potential of AMMs for engineering applications. Here, we review recent developments in AMMs with a focus on their future engineering, especially in the most promising fields of sound absorption/isolation, acoustic imaging, cloaking, and so on, furthermore, we outline the opportunities and challenges they are encountering.

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1. Introduction

Acoustics, a branch of physics concerning the generation, propagation, detection, and conversion of mechanical waves (i.e., sound and elastic waves), is playing an increasingly important role in modern engineering in the fields of communication, ultrasonic imaging, non-invasive high intensity focused ultrasound (HIFU) surgery, and more. In general, the manipulation of acoustic waves has always been a challenge to scientists and engineers because of the strong diffraction and penetration abilities of acoustic waves, which are due to their long wavelength and weak attenuation compared with other types of classical waves [1], despite the fact that strong penetration ability of acoustic waves can be an advantage in some applications, such as ultrasound imaging. With the advent of acoustic metamaterials (AMMs), the manipulation of complex sound and vibrations has become possible. AMMs can be dated back to the 1990s, with the emergence of sonic crystals (SCs) [2,3]. Through the construction of periodic individuals with high impedance contrast to the host matrices, it is possible to create SCs that exhibit unique energy band structures caused by strong Bragg scattering [4,5] of the prorogation waves. Furthermore, negative refraction [6,7], extraordinary transmission [8], and band edge states [9] have been observed in SCs.

With the discovery of metamaterials that have local resonance [10–12], the paradigm of AMMs has been further extended. Unlike SCs, which require the lattice parameters and the regulated wavelength to be on the same order, the lattice constant of local resonant AMMs is much smaller than the wavelength of the target acoustic waves. Thus, local resonant AMMs show potential advantages in the regulation of deep subwavelength waves. For a long time, researchers in diverse disciplines have designed various structures of AMMs with unprecedented constitutive parameters such as negative effective bulk modulus, negative effective mass density, and negative refractive index in order to achieve exotic functions, such as sound absorption [13], cloaking [14,15], focusing [16,17], and a reverse Doppler effect [18]. The related acoustic metasurfaces [19–22], space coiling AMMs [23–25], and miscellaneous AMMs have also greatly enriched the content of AMMs. In addition, inspired by quantum electronics and topological electronics, many AMMs have been designed to analogize these topological phenomena and properties, such as the Dirac-cone type of dispersion relation [26,27], pseudo-diffusion transport [28], topological one-way edge states [29–31], parity-time (PT)-symmetric acoustics [32–34], and non-reciprocal propagation [35,36].

This multifarious research, as mentioned above, has laid a solid foundation for engineering applications in the aviation industry, rail transit, biomedicine, the construction of new infrastructures,

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and more. However, the challenge of how to design and optimize AMMs to satisfy requirements in the real world remains to be overcome. To date, there are plenty of review articles and books [37–46] on AMMs, which provide a comprehensive and detailed exposition of their physical essence, specific categories of structures, and development context. Here, we will focus on the practical applications of AMMs by reviewing the significant progress that has been made in the fields of sound absorption/isolation, acoustic imaging, cloaking, and so forth. In addition, we will outline the opportunities and challenges of AMMs in future acoustic engineering applications.

2. Applications of acoustic metamaterials

2.1. Sound absorption/isolation

Noise pollution from engines, wheel/rail friction, air-conditioners, and so forth, is a major problem impacting human life in society today. Dissipating sound energy and isolating sound propagation are two strategies to reduce or even eliminate noise pollution. However, conventional materials for sound absorption/isolation are bulky [13,47] for low-frequency sound waves, due to their strong penetration and weak attenuation. By enhancing the energy density and employing the bandgap in resonant AMMs, subwavelength structures could efficiently improve sound absorption and block sound propagation, respectively. The vast majority of AMMs for sound absorption/isolation are constructed of decorated membrane resonators (DMRs), Helmholtz resonators (HRs), Fabry–Perot resonators (FPRs), and their miscellaneous variations.

The first DMR metamaterial, which has a circular membrane 0.28 mm thick and a small weight in its center, has achieved a low resonant frequency of 146 Hz [48]. The corresponding wavelength of 2.35 m is much longer than the structural size of the DMRs. Subsequently, thin elastic membranes decorated with designed patterns of rigid platelets [49] (Fig. 1(a)) were demonstrated. To overcome the geometric constraint, which leads to an intrinsic 50% limit for sound absorption [50], three possible strategies were proposed: One strategy is to place the DMRs close to a hard reflecting wall, as the multiple reflections between the wall and the DMRs can result in more than 99% energy absorption [51]. Utilizing the degenerate resonant frequencies of monopolar and dipolar resonators (Fig. 1(b)) is another strategy for perfect sound absorption [52]. The third strategy is based on the coherent perfect absorption (CPA) approach, which introduces a control wave in the opposite direction with the same amplitude but opposite phase from the target wave in order to eliminate the reflection and transmission, allowing 100% sound absorption to be realized [53–55]. In addition to noise absorption, DMR metamaterials can be used for noise isolation. Wang et al. [56] proposed a membrane-constrained type of DMR metamaterial that can suppress vibrations at the membrane center to isolate sound transmission at low frequencies. These unique characteristics of DMR metamaterials, along with their simple structure, small volume, and light weight, illustrate the potential applications of these metamaterials in reducing cabin noise in airliners, high-speed railways, automobiles, and ships.

Although DMRs show excellent sound absorption/isolation properties, it is easy for plastic films to fail after functioning for a long time, due to fatigue. Thus, using rigid solid materials to manufacture HRs and FPRs for sound absorption is another research focus. In general, a single HR exhibits monopolar resonances, resulting in a 50% absorption upper bound, due to similar geometric constraints as DMRs. To tackle this problem, Merkel et al. [57] demonstrated a perfect acoustic absorption through the interplay of the inherent losses and transparent modes; in addition, slightly over 50% (and near to 96%) one-sided absorption was achieved in symmetric (and asymmetric) structures. Furthermore, ventilation channels make it possible to absorb sound while maintaining fresh air circulation in a closed environment, such as a concert hall. In addition to the two-port case, an acoustic three-port network based on HRs has been proposed to operate as both a perfect absorber and a coherent perfect transmitter for the same frequency [58]. To further reduce the volume, the strategy of embedding apertures based on HRs [59,60] has been introduced.

The above examples confirm that increasing the energy density can greatly improve the sound absorption; however, the narrow bandwidth brought by the resonant features limits the absorption spectrum. A natural idea is to skillfully construct a hybrid system composed of multifrequency resonant units with a slight resonance frequency shift between one another to obtain an overall broad sound absorption spectrum [61–64]. In 2006, a theoretical method based on the radiation impedance of an array panel was proposed [65] to calculate the sound absorption of an HR array in a low-frequency range. Since then, this kind of narrow-neck HR has been widely explored to reduce the resonator's spatial scale and realize broadband sound absorption. One example is to insert one or more separate plates with small holes into the interior of an HR, leading to a perforated composite HR [66] (Fig. 1(c)), which can achieve continuous near-perfect absorption in a broadband range of 450–1360 Hz.

For FPRs, each channel requires one-quarter of a wavelength, which is a significant scale for absorption structures in the low-frequency range. Coiling or folding the channel is one efficient way to reduce the size of the absorbers. By stacking the channels together with various resonant frequencies, the coiled-up space structures [23,67,68] show good sound absorption performance in a broad bandwidth. However, the question of whether there exists a minimum thickness for an absorber with a given sound absorption performance over a designated frequency range remained unresolved until the discovery of the causality constraint [13,69], which revealed a tradeoff between minimizing the absorber's thickness and broadening its absorption spectrum. Under the guidance of the causality constraint, Yang et al. [69] realized a 10.86 cm-thick structure exhibiting a broadband, near-perfect flat absorption spectrum (Fig. 1(d)). In some cases, however, the causality constraint can be circumvented by manipulating the boundary condition on the backside of the absorber. Mak et al. [70] realized a minimum absorber thickness that was an order of magnitude less than the causal limit. Recently, based on the Fano-like interference, a planar acoustic ventilation barrier with a central hollow orifice and two surrounding helical pathways of varying pitch was proposed [71], which can block more than 90% of incident energy in the frequency range of 900–1418 Hz (Fig. 1(e)). Compared with DMR-type AMMs, HR and FPR metamaterials composed of rigid solid materials can greatly prolong the service life. Given their load-bearing and decorative functions, these AMMs hold promise for noise reduction in application such as expressways, smart power grids, and high-speed trains.

Waterborne sound absorption/isolation using AMMs is another direction for the potential application of AMMs in the fields of sonar shielding, underwater communication, ocean development, and so forth. However, research on waterborne sound absorption/isolation metamaterials progresses slowly because the high hydrostatic pressure and moderate impedance contrast hinder the construction of "meta-atoms." Some preliminary attempts have been based on embedded bubble structures [72–74] or high-density spheres coated by a viscoelastic matrix as resonant units [75]. One example involves AMMs based on patterned bubble arrays in water [76], which are inspired by biological superhydrophobic systems. This type of AMM can block sounds at low frequencies because of local resonances. By varying the sizes of the bubbles and the distance between them without changing the

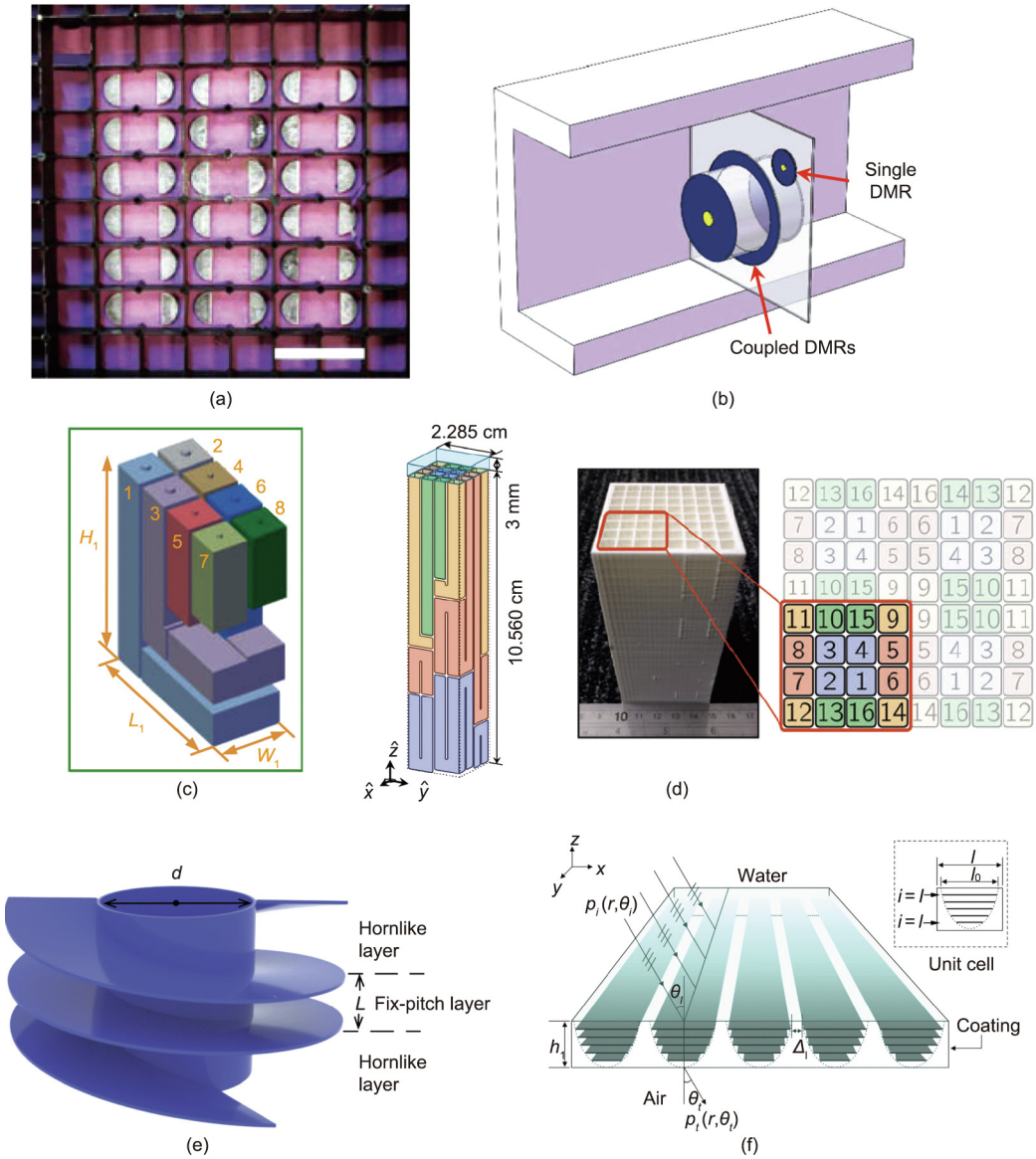

Fig. 1. AMMs for sound absorption/isolation. (a) Sample photograph of a dark AMM. (b) Schematic cutoff view of degenerate resonators. (c) View of a three-port network of HRs. (d) Schematic of perforated composite HRs. (e) Schematic of two spiral helices with a central orifice. (f) Schematic diagram of an absorber consisting of a viscoelastic coating layer embedded with periodically distributed plate scatterers. (a) Reproduced from Ref. [49] with permission; (b) reproduced from Ref. [52] with permission; (c) reproduced from Ref. [66] with permission; (d) reproduced from Ref. [69] with permission; (e) reproduced from Ref. [71] with permission; (f) reproduced from Ref. [78] with permission. The defination of all the abbreviations in figures could be find in the cited references.

thickness, the operating frequency can be tuned in the range of 9–1756 kHz. Duan et al. [77] reported another kind of lightweight AMM constructed of a rubber coating and an embedded metallic neck into the classical Helmholtz resonant cavity. With tailored internal geometric parameters, this type of quasi-Helmholtz resonator achieved almost perfect absorption from 306 to 921 Hz. To broaden the bandwidth of the sound absorption, Zhang et al. [78] recently proposed a method to use varied scatterer sizes (Fig. 1(f)), and Shi et al. [79] used multiple coating layers.

In addition to resonant structures, non-resonant metamaterials with a gradient index utilize the density change of the scatterers to modulate the local refractive index, so as to control the loss of incident sound waves. Based on the acoustic black hole principle and multiple scattering theory, Naify et al. [80] designed a cylindrical shell with a linearly graded sound velocity for omnidirectional underwater sound absorption. On this basis, Wang et al. [81] further developed a new two-dimensional (2D) axisymmetric underwater omnidirectional absorber with an acoustic gradient index structure and a hollow absorbing core. Nevertheless, the demonstration of megahertz ultrasonic waves corresponding to small wavelengths still presents difficulties for the processing and preparation of AMMs (see Section 3 for more detail).

2.2. Acoustic imaging

Acoustic imaging has important applications in medical ultrasound, nondestructive testing, and photoacoustic imaging. Pursuing high resolution has always been an important task in these fields. As with optical imaging, evanescent waves carry the detailed information of an object and exponentially decay as they leave the object's surface. As a result, conventional optical/acoustic imaging techniques often fail to collect the information from evanescent waves; therefore, the imaging resolution is strictly constrained by the diffraction limit. How to obtain the information carried by

evanescent waves is the key to realizing high-resolution imaging. Based on AMMs, two main strategies have been proposed. One relies on amplifying the evanescent waves for detection within a certain distance [82], and the other converts evanescent waves into propagating waves [83] for convenient measurements.

The first strategy is inspired by Pendry's [84] "perfect lens," which can strongly amplify optical evanescent waves by using negative refractive index materials [85,86]. In acoustics, a 2D membrane-based metamaterial has been designed that exhibits a negative refractive index. Using this AMM, a 17-fold net amplitude gain from an evanescent wave source has been achieved [78]. Similar designs have also been proposed for acoustic surface waves, where the smaller wavelength (compared with bulky acoustic waves) imposes challenges for imaging restricted by the diffraction limit [87]. It has been shown that, by using a 2D membrane-based metamaterial, the surface evanescent waves can be amplified and two-point sources with a distance of 1/17 wavelength can be resolved [88] (Fig. 2(a)). Beyond types of membrane, Kaina et al. [89] proposed a negative refractive index acoustic superlens composed of HRs (Fig. 2(b)), where subwavelength focusing seven times greater and a resolution 3.5 times better than the diffraction limit have been achieved.

The second strategy is based on anisotropic metamaterials that can convert evanescent waves into propagating waves due to their ability to couple large wave vector components (i.e., evanescent components). In general, the realization of this type of AMM is based on exciting trapped resonances inside acoustic waveguides [83,90,91]. By converting evanescent waves into propagating waves, anisotropic AMMs can realize high-resolution imaging even at the far field. For example, Li et al. [83] used 36 computer-numerical-controlled brass fins to convert evanescent wave components into propagating waves that can travel to the far field, where the associated deep subwavelength features remain intact (Fig. 2(c)).

In addition to the above-discussed strategies, by exciting waveguide resonances in a slab with a high-index contrast to its surrounding medium, the high spatial components of the subwavelength details can couple to the trapped modes of such a slab waveguide, and a subwavelength image can be obtained at a distant plane [92]. Super-oscillation acoustic wave packets generated by an ultrasonic metalens provide another means of breaking the diffraction limit for ultrasound imaging [93] (Fig. 2(d)). In regard to the narrow working bandwidth of AMMs, which can lead to serious chromatic aberration in imaging, some non-resonant elements [83] and a fin-shaped hyperlens [94–96] have been proposed to address this problem.

While most imaging applications of AMMs focus on airborne sound, some proposals on underwater acoustic lenses have begun to emerge [97,98]. For example, Peng et al. [99] designed gradient negative refractive index metamaterials to realize broadband underwater acoustic focusing, whose focal distance can be tuned by changing the lattice distance. These metamaterials can potentially be used in ultrasonic imaging, which is commonly based on the pulse-echo process that requires a broad bandwidth for focusing. Other methods to realize a broadband focusing lens with a gradient index involve utilizing 2D pentamode materials [100], using metasurfaces with non-resonant arrangements [101], or even using simple SCs made of single-phase rods, where an acoustic composite lens is realized to perform three-dimensional (3D) underwater focusing at a frequency range between 15 and 25 kHz [102] (Fig. 2(e)).

2.3. Other applications

In addition to the applications discussed above, there are several other noteworthy branches such as cloaking, unidirectional propagation, acoustic complementary metamaterials, and impedance matching. Cloaking, which was first proposed in optics [103,104], describes a phenomenon in which an object is invisible to incoming detection as if it disappears. This interesting phenomenon and its related subjects, such as illusion, have important applications in the fields of target stealth and security. Based on the transformation acoustic equations [105–107], various structures have been proposed for cloaking, such as alternating isotropic layers [108], multilayered composites [15], and pyramid-like perforated plate structures [109,110]. For underwater acoustic cloaking, Zhang et al. [111] proposed an AMM constructed with a planar network of subwavelength cavities connected by narrow channels (Fig. 3(a)).

Unidirectional sound propagation, in which the incident wave propagates only in one direction and is blocked in the other direction, behaves like a diode and can potentially be used for communication, anti-eavesdropping, and acoustic calculation. For such a non-reciprocal propagation, it is usually necessary to break the time-reversal symmetry by introducing a circulating fluid [36], which often leads to new problems such as instability and high viscosity. Another solution is based on breaking the spatial inversion symmetry, which gives rise to unidirectional sound propagation with higher efficiency, broader bandwidth, and much lower power consumption [112] (Fig. 3(b)).

To realize unidirectional propagation or to reduce backscattering, another flourishing field has recently attracted a great deal of attention—namely, the topological phases of matter [113,114]. Originating in electronic systems and firstly found to be responsible for the famous integer quantum Hall effect, topological physics was soon extended into the classical realms of optics [115,116] and acoustics [117,118], where metamaterials serve as resourceful material platforms to greatly enrich and simultaneously benefit from studies of the topological phases of matter. Due to their nature of topological protection, topological materials support robust interface states that are immune to backscattering [117]. This feature enables revolutionary applications in various fields, including low-loss waveguides, topological filtering, and collimations. More interestingly, topological phases of matter bring new degrees of freedom to light and sound control. For example, by designing a 2D honeycomb lattice composed of steel rods, He et al. [119] realized the acoustic analog of the quantum spin Hall effect, which supports backscattering-immune sound transportation (Fig. 3(c)). The creation of pseudospins in acoustics opens up the field of acoustic spintronics [22,120], which may have interesting applications in acoustic communications and spin-dependent information processing. In addition, recent developments in acoustic valley insulators [121], Weyl semi-metals [122], and higher-order topological insulators [31] has further inspired various methods for sound control, in addition to versatile applications.

Moreover, when sound waves encounter obstacles or interfaces with a large impedance mismatch, a considerable amount of acoustic energy will be scattered. To restore the distorted acoustic fields, acoustic complementary metamaterials (CMMs) have been proposed. In general, CMMs simultaneously require negative and anisotropic material properties in order to prevent large resonant damping. They have potential applications in transcranial ultrasound and nondestructive testing, where sound waves often encounter aberrating layers such as skulls, steel plates, and layers with a large impedance mismatch. For example, a quasi-2D acoustic CMM utilizing clamped membranes and side branches [123] has been proposed to restore the diffraction from an aberrating layer in a noninvasive manner (Fig. 3(d)).

Overcoming the large impedance mismatch and enhancing sound transmission at the water–air interface hold great significance for underwater communication, imaging, and remote sensing. Acoustic metasurfaces have been demonstrated to be able to

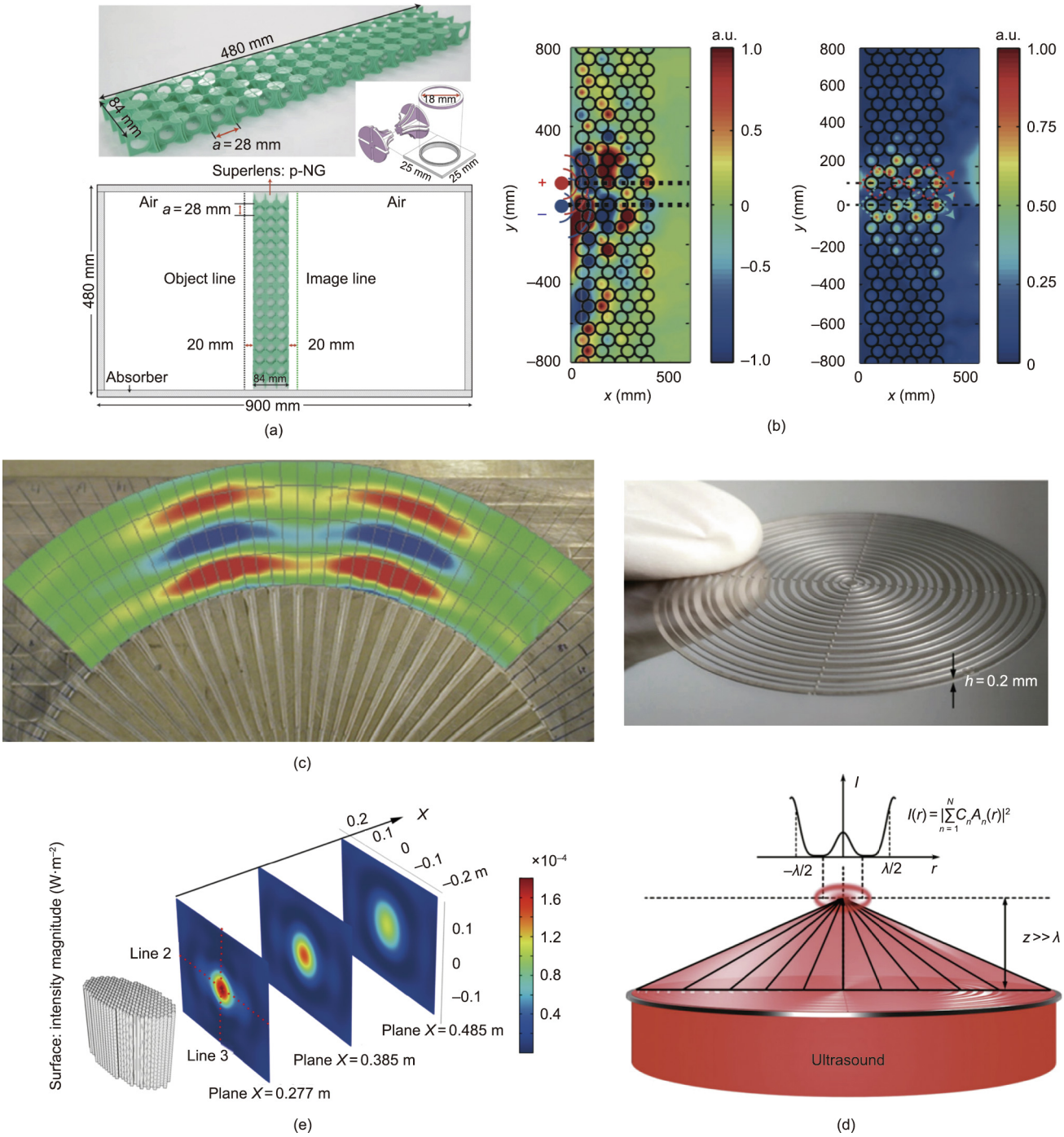

Fig. 2. AMMs for imaging. (a) Superlensing slab made of a membrane-based AMM. (b) Super-resolution imaging for two sources out of phase by using a flat acoustic superlens. (c) Experimental demonstration of the magnifying imaging of a sub-diffraction-limited dual-source. (d) Photograph of an ultrasonic metalens (upper panel) and a super-oscillating field pattern beyond the evanescent region (lower panel). (e) Numerical results of the wave intensity for an approximate plane wave. (a) Reproduced from Ref. [88] with permission; (b) reproduced from Ref. [89] with permission; (c) reproduced from Ref. [83] with permission; (d) reproduced from Ref. [93] with permission; (e) reproduced from Ref. [102] with permission. The defination of all the abbreviations in figures could be find in the cited references.

break through the limitation of quarter-wave plates or multilayered antireflection coatings for impedance matching [124]. A metasurface consisting of a locked air layer and a hydrophobic solid substrate with multiple hollow cells has been proposed to enhance the energy transmission by over 200 times at the frequency range of 10–4000 Hz [125]. By changing the immersion depth of the solid substrate, the working frequency can be tuned while maintaining the energy transmission. Inspired by the "lotus effect," a microscale air layer is constructed using a superhydrophobic aluminum plate and water [126]. A nearly 100% energy transmission coefficient and a wide-angle water-to-air acoustic transmission can be obtained in the frequency range of 10–100 kHz.

Another application is in medical ultrasound, which requires impedance matching between piezoelectric materials and human body tissues. Introducing an AMM layer with a gradient index is a common means of achieving impedance matching [127]. Li

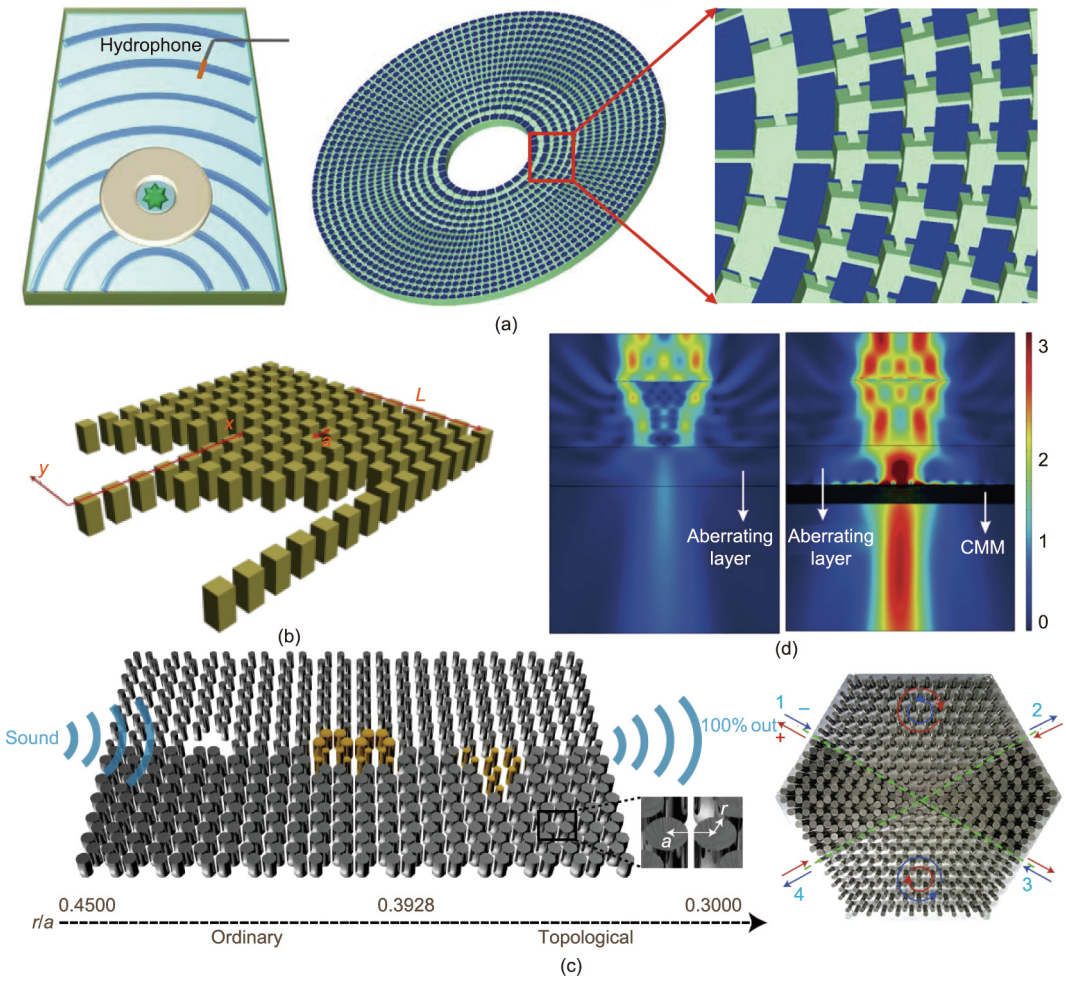

Fig. 3. (a) Schematic diagram of acoustic cloaking and the configuration of an acoustic cylindrical cloak. (b) Schematic diagram of a sonic-crystal-based acoustic diode with a periodic arrangement in the *y*-direction. (c) Schematic of an acoustic topological insulator constructed with two types of phononic "graphene" with the same lattice constant but different "atom" (i.e., stainless-steel rod) radii (left) and a photo of the cross-waveguide splitter (right). (d) Acoustic intensity fields of an unfocused beam in a skull only (left) and with CMM and the skull (right). (a) Reproduced from Ref. [111] with permission; (b) reproduced from Ref. [112] with permission; (c) reproduced from Ref. [119] with permission; (d) reproduced from Ref. [123] with permission. The defination of all the abbreviations in figures could be find in the cited references.

et al. [128] proposed an anisotropic cone-structured AMM matching layer with a gradient acoustic impedance along the ultrasound propagation direction to realize the extraordinary transmission of ultrasound over a wide frequency range. In the abovementioned studies, the impedance-matching effect is efficient for normal incidence but gradually deteriorates when the incident angle increases. To achieve wide-angle acoustic impedance matching, Liu et al. [129] used silicone rubber to design one-dimensional (1D) and 2D acoustic structures by applying the effective medium theory with spatially dispersive parameters. In this way, wide-angle and broadband impedance matching with water was realized, leading to near 100% transmission of acoustic waves in water.

3. Discussion

In this perspective, we first briefly provided a panorama of AMMs to establish a framework for this topic. We then reviewed the potential applications of AMMs in several promising fields, such as sound absorption/isolation, imaging, acoustics, and cloaking. Despite the great success of fundamental research on AMMs, AMMs still present several challenges in practical engineering.

The narrow working bandwidth induced by resonances and the inevitable energy loss resulting in low transmission coefficients can substantially impede the practical application of AMMs. The two main solutions for the former issue are integrating AMMs with a broad resonant spectrum and designing tunable active metamaterials. As a possible solution for the latter issue, the recent development of topological insulators and associated topological devices provides robust lossless acoustic propagation, thanks to their topological protection. In addition, loudspeakers, transducers, and other energy-conversion methods (e.g., multi-physics coupling with temperature or electromagnetic regulations) can be employed to compensate for the energy dissipation of AMMs. Other explorations of, for example, the physics of non-Hermiticity and the parity-time symmetric properties of AMMs may reveal even more novel and interesting ways to manipulate acoustic waves and facilitate their application accordingly.

Aside from airborne AMMs, growing attention is being paid to the design of metamaterials used in fluids such as water or in human tissues. However, as discussed above, the impedance contrast between fluids and solid AMMs is relatively small (only about one order of magnitude). A considerable amount of acoustic energy transmitting into both fluids and solid structures could result in a complex travel path and therefore limit the obtainable effective

acoustic parameters. One strategy that might address this challenge is to build air bubbles as resonators or to wrap air with elastic film bubbles; in such contexts, the effective parameters can be modulated by varying the preset pressure of the bubbles or by choosing different wrapping materials.

Another challenge lies in the manufacturing of AMMs. Simple-structure AMMs [130,131] can be fabricated by means of conventional fabrication approaches, including machining, casting, injection molding, and laser cutting. AMMs with complicated 3D structures, such as channels [67], chambers [132], and spherical or cylindrical fillers [10], mainly rely on manual manufacturing and 3D printing. However, massive manual manufacturing potentially leads to a loss of consistency and unstable product quality. 3D printing technology is well-suited to manufacture complicated structures. Nevertheless, when it comes to massive production, 3D printing may suffer from high-cost consumption, low production efficiency, and limited varieties of materials. Therefore, on the route to real-world AMM applications, it is particularly urgent to develop a new generation of 3D printing technology with low cost, high efficiency, and various materials. Furthermore, AMMs in the megahertz range are of great value for improving the resolution of medical ultrasonic imaging and the focusing of high-precision noninvasive surgeries. However, manufacturing methods at the micron scale are rarely reported. Thus, advanced micro/nanofabrication is urgently demanded in the preparation of high-frequency metamaterials.

In addition, emerging ideas on controlling sound fields globally and tailoring AMMs for expected functions and properties have become essential for the use of AMMs in practical engineering, which must be compatible with existing acoustic devices. However, determining the complex relationship between the structures, functions, and properties of AMMs requires considerable effort. Digitizing AMMs in combination with artificial intelligence is the most likely path to save time and effort on non-innovative designing. As shown in Fig. 4, an overall consideration of fundamentals, performance, fabrication, and application in the design and optimization of exquisite AMMs that are balanced in all aspects is the key to ultimate industrial applications.

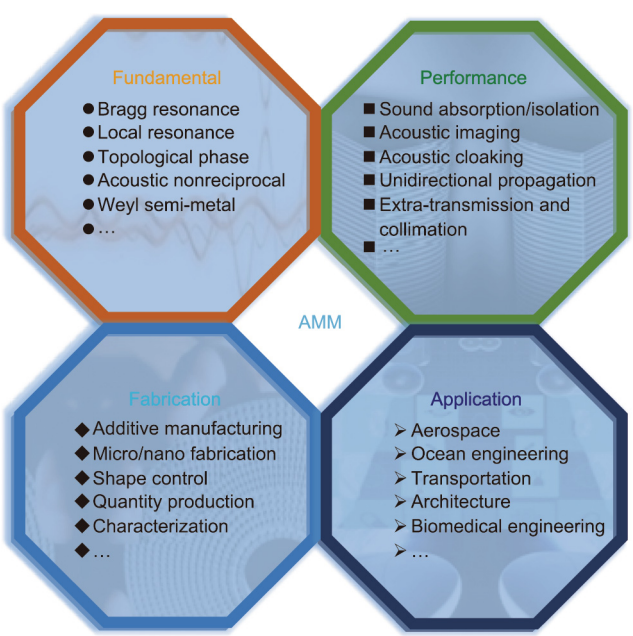

Fig. 4. Factors should be comprehensively considered in the design and preparation of AMMs.

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Compliance with ethics guidelines

Qiangbing Lu, Xin Li, Xiujuan Zhang, Minghui Lu, and Yanfeng Chen declare that they have no conflict of interest or financial conflicts to disclose.

References

[1] Kinsler LE, Frey AR, Coppens AB, Sanders JV. Fundamentals of acoustics. 4th ed. Hoboken: John Wiley & Sons, Inc.; 2000.
[2] Schriemer HP, Cowan ML, Page JH, Sheng P, Liu Z, Weitz DA. Energy velocity of diffusing waves in strongly scattering media. Phys Rev Lett 1997;79(17):3166–9.
[3] Kushwaha MS, Halevi P, Dobrzynski L, Djafari-Rouhani B. Acoustic band structure of periodic elastic composites. Phys Rev Lett 1993;71(13):2022–5.
[4] Sigalas M, Economou EN. Band structure of elastic waves in two dimensional systems. Solid State Commun 1993;86(3):141–3.
[5] Martínez-Sala R, Sancho J, Sánchez JV, Gómez V, Llinares J, Meseguer F. Sound attenuation by sculpture. Nature 1995;378(6554):241.
[6] Yang S, Page JH, Liu Z, Cowan ML, Chan CT, Sheng P. Focusing of sound in a 3D phononic crystal. Phys Rev Lett 2004;93(2):024301.
[7] Ke M, Liu Z, Qiu C, Wang W, Shi J, Wen W, et al. Negative-refraction imaging with two-dimensional phononic crystals. Phys Rev B 2005;72(6):064306.
[8] Lu MH, Liu XK, Feng L, Li J, Huang CP, Chen YF, et al. Extraordinary acoustic transmission through a 1D grating with very narrow apertures. Phys Rev Lett 2007;99(17):174301.
[9] Qiu C, Liu Z. Acoustic directional radiation and enhancement caused by band-edge states of two-dimensional phononic crystals. Appl Phys Lett 2006;89(6):063106.
[10] Liu Z, Zhang X, Mao Y, Zhu YY, Yang Z, Chan CT, et al. Locally resonant sonic materials. Science 2000;289(5485):1734–6.
[11] Ho KM, Cheng CK, Yang Z, Zhang XX, Sheng P. Broadband locally resonant sonic shields. Appl Phys Lett 2003;83(26):5566–8.
[12] Sainidou R, Djafari-Rouhani B, Pennec Y, Vasseur JO. Locally resonant phononic crystals made of hollow spheres or cylinders. Phys Rev B 2006;73(2):024302.
[13] Yang M, Sheng P. Sound absorption structures: from porous media to acoustic metamaterials. Annu Rev Mater Res 2017;47(1):83–114.
[14] Farhat M, Enoch S, Guenneau S, Movchan AB. Broadband cylindrical acoustic cloak for linear surface waves in a fluid. Phys Rev Lett 2008;101(13):134501.
[15] Torrent D, Sánchez-Dehesa J. Acoustic cloaking in two dimensions: a feasible approach. New J Phys 2008;10(6):063015.
[16] Zhang S, Yin L, Fang N. Focusing ultrasound with an acoustic metamaterial network. Phys Rev Lett 2009;102(19):194301.
[17] Li Y, Yu G, Liang B, Zou X, Li G, Cheng S, et al. Three-dimensional ultrathin planar lenses by acoustic metamaterials. Sci Rep 2014;4(1):6830.
[18] Lee SH, Park CM, Seo YM, Kim CK. Reversed Doppler effect in double negative metamaterials. Phys Rev B 2010;81(24):241102.
[19] Li Y, Jiang X, Liang B, Cheng J, Zhang L. Metascreen-based acoustic passive phased array. Phys Rev Appl 2015;4(2):024003.
[20] Cheng Y, Zhou C, Yuan BG, Wu DJ, Wei Q, Liu XJ. Ultra-sparse metasurface for high reflection of low-frequency sound based on artificial Mie resonances. Nat Mater 2015;14(10):1013–9.
[21] Xie Y, Wang W, Chen H, Konneker A, Popa BI, Cummer SA. Wavefront modulation and subwavelength diffractive acoustics with an acoustic metasurface. Nat Commun 2014;5(1):5553.
[22] Ge H, Xu XY, Liu L, Xu R, Lin ZK, Yu SY, et al. Observation of acoustic skyrmions. Phys Rev Lett 2021;127(14):144502.
[23] Wang Y, Zhao H, Yang H, Zhong J, Zhao D, Lu Z, et al. A tunable sound-absorbing metamaterial based on coiled-up space. J Appl Phys 2018;123(18):185109.
[24] Song GY, Cheng Q, Huang B, Dong HY, Cui TJ. Broadband fractal acoustic metamaterials for low-frequency sound attenuation. Appl Phys Lett 2016;109(13):131901.
[25] Liang Z, Li J. Extreme acoustic metamaterial by coiling up space. Phys Rev Lett 2012;108(11):114301.
[26] Yu SY, Sun XC, Ni X, Wang Q, Yan XJ, He C, et al. Surface phononic graphene. Nat Mater 2016;15(12):1243–7.
[27] Torrent D, Sánchez-Dehesa J. Acoustic analogue of graphene: observation of Dirac cones in acoustic surface waves. Phys Rev Lett 2012;108(17):174301.
[28] Zhang X, Liu Z. Extremal transmission and beating effect of acoustic waves in two-dimensional sonic crystals. Phys Rev Lett 2008;101(26):264303.
[29] Chen ZG, Wu Y. Tunable topological phononic crystals. Phys Rev Appl 2016;5(5):054021.

[30] Ni X, He C, Sun XC, Liu X, Lu MH, Feng L, et al. Topologically protected one-way edge mode in networks of acoustic resonators with circulating air flow. New J Phys 2015;17(5):053016.
[31] Zhang X, Wang HX, Lin ZK, Tian Y, Xie B, Lu MH, et al. Second-order topology and multidimensional topological transitions in sonic crystals. Nat Phys 2019;15(6):582–8.
[32] Christensen J, Willatzen M, Velasco VR, Lu MH. Parity-time synthetic phononic media. Phys Rev Lett 2016;116(20):207601.
[33] Zhu X, Ramezani H, Shi C, Zhu J, Zhang X. PT-symmetric acoustics. Phys Rev X 2014;4(3):031042.
[34] Hu B, Zhang Z, Zhang H, Zheng L, Xiong W, Yue Z, et al. Non-Hermitian topological whispering gallery. Nature 2021;597(7878):655–9.
[35] Popa BI, Cummer SA. Non-reciprocal and highly nonlinear active acoustic metamaterials. Nat Commun 2014;5(1):3398.
[36] Fleury R, Sounas DL, Sieck CF, Haberman MR, Alù A. Sound isolation and giant linear nonreciprocity in a compact acoustic circulator. Science 2014;343(6170):516–9.
[37] Liao G, Luan C, Wang Z, Liu J, Yao X, Fu J. Acoustic metamaterials: a review of theories, structures, fabrication approaches, and applications. Adv Mater Technol 2021;6(5):2000787.
[38] Zangeneh-Nejad F, Fleury R. Active times for acoustic metamaterials. Rev Phys 2019;4:100031.
[39] Wu Y, Yang M, Sheng P. Perspective: acoustic metamaterials in transition. J Appl Phys 2018;123(9):090901.
[40] Ge H, Yang M, Ma C, Lu MH, Chen YF, Fang N, et al. Breaking the barriers: advances in acoustic functional materials. Natl Sci Rev 2018;5(2):159–82.
[41] Cummer SA, Christensen J, Alù A. Controlling sound with acoustic metamaterials. Nat Rev Mater 2016;1(3):16001.
[42] Lu MH, Feng L, Chen YF. Phononic crystals and acoustic metamaterials. Mater Today 2009;12(12):34–42.
[43] Muhammad LCW. From photonic crystals to seismic metamaterials: a review via phononic crystals and acoustic metamaterials. Arch Comput Methods Eng 2021;29:1137–98.
[44] Kumar S, Lee HP. The present and future role of acoustic metamaterials for architectural and urban noise mitigations. Acoustics 2019;1(3):590–607.
[45] Liu J, Guo H, Wang T. A review of acoustic metamaterials and phononic crystals. Crystals 2020;10(4):305.
[46] Kumar S, Lee HP. Recent advances in active acoustic metamaterials. Int J Appl Mech 2019;11(8):1950081.
[47] Allard J, Atalla N. Propagation of sound in porous media: modelling sound absorbing materials. 2nd ed. Chichester: John Wiley & Sons, Ltd.; 2009.
[48] Yang Z, Mei J, Yang M, Chan NH, Sheng P. Membrane-type acoustic metamaterial with negative dynamic mass. Phys Rev Lett 2008;101(20):204301.
[49] Mei J, Ma G, Yang M, Yang Z, Wen W, Sheng P. Dark acoustic metamaterials as super absorbers for low-frequency sound. Nat Commun 2012;3(1):756.
[50] Yang M, Li Y, Meng C, Fu C, Mei J, Yang Z, et al. Sound absorption by subwavelength membrane structures: a geometric perspective. CR Mecanique 2015;343(12):635–44.
[51] Ma G, Yang M, Xiao S, Yang Z, Sheng P. Acoustic metasurface with hybrid resonances. Nat Mater 2014;13(9):873–8.
[52] Yang M, Meng C, Fu C, Li Y, Yang Z, Sheng P. Subwavelength total acoustic absorption with degenerate resonators. Appl Phys Lett 2015;107(10):104104.
[53] Wei P, Croënne C, Tak Chu S, Li J. Symmetrical and anti-symmetrical coherent perfect absorption for acoustic waves. Appl Phys Lett 2014;104(12):121902.
[54] Yang M, Ma G, Yang Z, Sheng P. Subwavelength perfect acoustic absorption in membrane-type metamaterials: a geometric perspective. EPJ Appl Metamat 2015;2:10.
[55] Meng C, Zhang X, Tang ST, Yang M, Yang Z. Acoustic coherent perfect absorbers as sensitive null detectors. Sci Rep 2017;7(1):43574.
[56] Wang X, Zhao H, Luo X, Huang Z. Membrane-constrained acoustic metamaterials for low frequency sound insulation. Appl Phys Lett 2016;108(4):041905.
[57] Merkel A, Theocharis G, Richoux O, Romero-García V, Pagneux V. Control of acoustic absorption in one-dimensional scattering by resonant scatterers. Appl Phys Lett 2015;107(24):244102.
[58] Richoux O, Achilleos V, Theocharis G, Brouzos I. Subwavelength interferometric control of absorption in three-port acoustic network. Sci Rep 2018;8(1):12328.
[59] Groby JP, Lagarrigue C, Brouard B, Dazel O, Tournat V, Nennig B. Enhancing the absorption properties of acoustic porous plates by periodically embedding Helmholtz resonators. J Acoust Soc Am 2015;137(1):273–80.
[60] Huang S, Fang X, Wang X, Assouar B, Cheng Q, Li Y. Acoustic perfect absorbers via Helmholtz resonators with embedded apertures. J Acoust Soc Am 2019;145(1):254–62.
[61] Jiménez N, Romero-García V, Pagneux V, Groby JP. Quasiperfect absorption by subwavelength acoustic panels in transmission using accumulation of resonances due to slow sound. Phys Rev B 2017;95(1):014205.
[62] Groby JP, Pommier R, Aurégan Y. Use of slow sound to design perfect and broadband passive sound absorbing materials. J Acoust Soc Am 2016;139(4):1660–71.
[63] Jiang X, Liang B, Li R, Zou X, Yin L, Cheng J. Ultra-broadband absorption by acoustic metamaterials. Appl Phys Lett 2014;105(24):243505.
[64] Romero-García V, Theocharis G, Richoux O, Pagneux V. Use of complex frequency plane to design broadband and sub-wavelength absorbers. J Acoust Soc Am 2016;139(6):3395–403.
[65] Kim SR, Kim YH, Jang JH. A theoretical model to predict the low-frequency sound absorption of a helmholtz resonator array. J Acoust Soc Am 2006;119(4):1933–6.
[66] Liu CR, Wu JH, Chen X, Ma F. A thin low-frequency broadband metasurface with multi-order sound absorption. J Phys D Appl Phys 2019;52(10):105302.
[67] Shen Y, Yang Y, Guo X, Shen Y, Zhang D. Low-frequency anechoic metasurface based on coiled channel of gradient cross-section. Appl Phys Lett 2019;114(8):083501.
[68] Zhang C, Hu X. Three-dimensional single-port labyrinthine acoustic metamaterial: perfect absorption with large bandwidth and tunability. Phys Rev Appl 2016;6(6):064025.
[69] Yang M, Chen S, Fu C, Sheng P. Optimal sound-absorbing structures. Mater Horiz 2017;4(4):673–80.
[70] Mak HY, Zhang X, Dong Z, Miura S, Iwata T, Sheng P. Going beyond the causal limit in acoustic absorption. Phys Rev Appl 2021;16(4):044062.
[71] Sun M, Fang X, Mao D, Wang X, Li Y. Broadband acoustic ventilation barriers. Phys Rev Appl 2020;13(4):044028.
[72] Leroy V, Strybulevych A, Lanoy M, Lemoult F, Tourin A, Page JH. Superabsorption of acoustic waves with bubble metascreens. Phys Rev B 2015;91(2):020301.
[73] Ivansson SM. Sound absorption by viscoelastic coatings with periodically distributed cavities. J Acoust Soc Am 2006;119(6):3558–67.
[74] Ivansson SM. Numerical design of Alberich anechoic coatings with superellipsoidal cavities of mixed sizes. J Acoust Soc Am 2008;124(4):1974–84.
[75] Meng H, Wen J, Zhao H, Wen X. Optimization of locally resonant acoustic metamaterials on underwater sound absorption characteristics. J Sound Vibrat 2012;331(20):4406–16.
[76] Huang Z, Zhao S, Su M, Yang Q, Li Z, Cai Z, et al. Bioinspired patterned bubbles for broad and low-frequency acoustic blocking. ACS Appl Mater Interfaces 2020;12(1):1757–64.
[77] Duan M, Yu C, Xin F, Lu TJ. Tunable underwater acoustic metamaterials via quasi-Helmholtz resonance: from low-frequency to ultra-broadband. Appl Phys Lett 2021;118(7):071904.
[78] Zhang Y, Pan J, Chen K, Zhong J. Subwavelength and quasi-perfect underwater sound absorber for multiple and broad frequency bands. J Acoust Soc Am 2018;144(2):648–59.
[79] Shi K, Jin G, Liu R, Ye T, Xue Y. Underwater sound absorption performance of acoustic metamaterials with multilayered locally resonant scatterers. Results Phys 2019;12:132–42.
[80] Naify CJ, Martin TP, Layman CN, Nicholas M, Thangawng AL, Calvo DC, et al. Underwater acoustic omnidirectional absorber. Appl Phys Lett 2014;104(7):073505.
[81] Wang C, Li SD, Zheng WG, Huang QB. Acoustic absorption characteristics of new underwater omnidirectional absorber. Chin Phys Lett 2019;36(4):044301.
[82] Park CM, Park JJ, Lee SH, Seo YM, Kim CK, Lee SH. Amplification of acoustic evanescent waves using metamaterial slabs. Phys Rev Lett 2011;107(19):194301.
[83] Li J, Fok L, Yin X, Bartal G, Zhang X. Experimental demonstration of an acoustic magnifying hyperlens. Nat Mater 2009;8(12):931–4.
[84] Pendry JB. Negative refraction makes a perfect lens. Phys Rev Lett 2000;85(18):3966–9.
[85] Zhang X, Liu Z. Superlenses to overcome the diffraction limit. Nat Mater 2008;7(6):435–41.
[86] Fang N, Lee H, Sun C, Zhang X. Sub-diffraction-limited optical imaging with a silver superlens. Science 2005;308(5721):534–7.
[87] Ambati M, Fang N, Sun C, Zhang X. Surface resonant states and superlensing in acoustic metamaterials. Phys Rev B 2007;75(19):195447.
[88] Park JJ, Park CM, Lee KJB, Lee SH. Acoustic superlens using membrane-based metamaterials. Appl Phys Lett 2015;106(5):051901.
[89] Kaina N, Lemoult F, Fink M, Lerosey G. Negative refractive index and acoustic superlens from multiple scattering in single negative metamaterials. Nature 2015;525(7567):77–81.
[90] Molerón M, Daraio C. Acoustic metamaterial for subwavelength edge detection. Nat Commun 2015;6(1):8037.
[91] Ao X, Chan CT. Far-field image magnification for acoustic waves using anisotropic acoustic metamaterials. Phys Rev E 2008;77(2):025601.
[92] Christensen J, García de Abajo FJ. Acoustic field enhancement and subwavelength imaging by coupling to slab waveguide modes. Appl Phys Lett 2010;97(16):164103.
[93] Shen YX, Peng YG, Cai F, Huang K, Zhao DG, Qiu CW, et al. Ultrasonic super-oscillation wave-packets with an acoustic meta-lens. Nat Commun 2019;10(1):3411.
[94] García-Chocano VM, Christensen J, Sánchez-Dehesa J. Negative refraction and energy funneling by hyperbolic materials: an experimental demonstration in acoustics. Phys Rev Lett 2014;112(14):144301.
[95] Shen C, Xie Y, Sui N, Wang W, Cummer SA, Jing Y. Broadband acoustic hyperbolic metamaterial. Phys Rev Lett 2015;115(25):254301.
[96] Lemoult F, Fink M, Lerosey G. Acoustic resonators for far-field control of sound on a subwavelength scale. Phys Rev Lett 2011;107(6):064301.
[97] Bai L, Dong HY, Song GY, Cheng Q, Huang B, Jiang WX, et al. Impedance-matching wavefront-transformation lens based on acoustic metamaterials. Adv Mater Technol 2018;3(11):1800064.
[98] Al Jahdali R, Wu Y. High transmission acoustic focusing by impedance-matched acoustic meta-surfaces. Appl Phys Lett 2016;108(3):031902.

[99] Peng S, He Z, Jia H, Zhang A, Qiu C, Ke M, et al. Acoustic far-field focusing effect for two-dimensional graded negative refractive-index sonic crystals. Appl Phys Lett 2010;96(26):263502.
[100] Su X, Norris AN, Cushing CW, Haberman MR, Wilson PS. Broadband focusing of underwater sound using a transparent pentamode lens. J Acoust Soc Am 2017;141(6):4408–17.
[101] Chen J, Rao J, Lisevych D, Fan Z. Broadband ultrasonic focusing in water with an ultra-compact metasurface lens. Appl Phys Lett 2019;114(10):104101.
[102] Ruan Y, Liang X, Wang Z, Wang T, Deng Y, Qu F, et al. 3-D underwater acoustic wave focusing by periodic structure. Appl Phys Lett 2019;114(8):081908.
[103] Schurig D, Mock JJ, Justice BJ, Cummer SA, Pendry JB, Starr AF, et al. Metamaterial electromagnetic cloak at microwave frequencies. Science 2006;314(5801):977–80.
[104] Pendry JB, Schurig D, Smith DR. Controlling electromagnetic fields. Science 2006;312(5781):1780–2.
[105] Cummer SA, Popa BI, Schurig D, Smith DR, Pendry J, Rahm M, et al. Scattering theory derivation of a 3D acoustic cloaking shell. Phys Rev Lett 2008;100(2):024301.
[106] Chen H, Chan C. Acoustic cloaking in three dimensions using acoustic metamaterials. Appl Phys Lett 2007;91(18):183518.
[107] Cummer SA, Schurig D. One path to acoustic cloaking. New J Phys 2007;9(3):45.
[108] Cheng Y, Yang F, Xu JY, Liu XJ. A multilayer structured acoustic cloak with homogeneous isotropic materials. Appl Phys Lett 2008;92(15):151913.
[109] Popa BI, Zigoneanu L, Cummer SA. Experimental acoustic ground cloak in air. Phys Rev Lett 2011;106(25):253901.
[110] Zigoneanu L, Popa BI, Cummer SA. Three-dimensional broadband omnidirectional acoustic ground cloak. Nat Mater 2014;13(4):352–5.
[111] Zhang S, Xia C, Fang N. Broadband acoustic cloak for ultrasound waves. Phys Rev Lett 2011;106(2):024301.
[112] Li XF, Ni X, Feng L, Lu MH, He C, Chen YF. Tunable unidirectional sound propagation through a sonic-crystal-based acoustic diode. Phys Rev Lett 2011;106(8):084301.
[113] Hasan MZ, Kane CL. Colloquium: topological insulators. Rev Mod Phys 2010;82(4):3045–67.
[114] Qi XL, Zhang SC. Topological insulators and superconductors. Rev Mod Phys 2011;83(4):1057–110.
[115] Zhang L, Ren J, Wang JS, Li B. Topological nature of the phonon Hall effect. Phys Rev Lett 2010;105(22):225901.
[116] Li N, Ren J, Wang L, Zhang G, Hänggi P, Li B. Colloquium: phononics: manipulating heat flow with electronic analogs and beyond. Rev Mod Phys 2012;84(3):1045–66.
[117] Yang Z, Gao F, Shi X, Lin X, Gao Z, Chong Y, et al. Topological acoustics. Phys Rev Lett 2015;114(11):114301.
[118] Xiao M, Ma G, Yang Z, Sheng P, Zhang ZQ, Chan CT. Geometric phase and band inversion in periodic acoustic systems. Nat Phys 2015;11(3):240–4.
[119] He C, Ni X, Ge H, Sun XC, Chen YB, Lu MH, et al. Acoustic topological insulator and robust one-way sound transport. Nat Phys 2016;12(12):1124–9.
[120] Zhang Z, Wei Q, Cheng Y, Zhang T, Wu D, Liu X. Topological creation of acoustic pseudospin multipoles in a flow-free symmetry-broken metamaterial lattice. Phys Rev Lett 2017;118(8):084303.
[121] Zhu Z, Yan M, Pan J, Yang Y, Deng W, Lu J, et al. Acoustic valley spin Chern insulators. Phys Rev Appl 2021;16(1):014058.
[122] Yang Z, Zhang B. Acoustic type-II Weyl nodes from stacking dimerized chains. Phys Rev Lett 2016;117(22):224301.
[123] Shen C, Xu J, Fang NX, Jing Y. Anisotropic complementary acoustic metamaterial for canceling out aberrating layers. Phys Rev X 2014;4(4):041033.
[124] Bok E, Park JJ, Choi H, Han CK, Wright OB, Lee SH. Metasurface for water-to-air sound transmission. Phys Rev Lett 2018;120(4):044302.
[125] Huang Z, Zhao S, Zhang Y, Cai Z, Li Z, Xiao J, et al. Tunable fluid-type metasurface for wide-angle and multifrequency water–air acoustic transmission. Research 2021;2021:9757943.
[126] Huang Z, Zhao Z, Zhao S, Cai X, Zhang Y, Cai Z, et al. Lotus metasurface for wide-angle intermediate-frequency water–air acoustic transmission. ACS Appl Mater Interfaces 2021;13(44):53242–51.
[127] Ding Y, Statharas EC, Yao K, Hong M. A broadband acoustic metamaterial with impedance matching layer of gradient index. Appl Phys Lett 2017;110(24):241903.
[128] Li Z, Yang DQ, Liu SL, Yu SY, Lu MH, Zhu J, et al. Broadband gradient impedance matching using an acoustic metamaterial for ultrasonic transducers. Sci Rep 2017;7(1):42863.
[129] Liu C, Luo J, Lai Y. Acoustic metamaterials with broadband and wide-angle impedance matching. Phys Rev Mater 2018;2(4):045201.
[130] Fernández-Marín AA, Jiménez N, Groby JP, Sánchez-Dehesa J, Romero-García V. Aerogel-based metasurfaces for perfect acoustic energy absorption. Appl Phys Lett 2019;115(6):061901.
[131] Song K, Kim J, Hur S, Kwak JH, Lee SH, Kim T. Directional reflective surface formed via gradient-impeding acoustic meta-surfaces. Sci Rep 2016;6(1):32300.
[132] Xie Y, Popa BI, Zigoneanu L, Cummer SA. Measurement of a broadband negative index with space-coiling acoustic metamaterials. Phys Rev Lett 2013;110(17):175501.